WEEK 1 REPORT

Lab Setup, Linux and Networking Foundations

Cybersecurity Internship Weekly Report

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1. Introduction

During Week 1 of the DG Interns Hub internship, I focused on understanding the fundamental concepts of cybersecurity, Linux operating systems, computer networking, and virtualisation.

I studied cybersecurity theory and practised Linux commands in a laboratory environment. I also worked through beginner-level cybersecurity exercises on TryHackMe and OverTheWire Bandit to improve my technical knowledge and command-line skills.

This report summarises the topics studied, practical activities performed, tools used, learning outcomes, and evidence collected during Week 1.

2. Objectives

The main objectives of Week 1 were:

- Understand the fundamentals of cybersecurity and the CIA triad.
- Learn the differences between SOC analysts, penetration testers, and incident responders.
- Understand the OSI and TCP/IP networking models.
- Learn about IP addresses, MAC addresses, DNS, DHCP, ports, and common protocols.
- Practise Linux commands, file operations, permissions, users, groups, and scheduled tasks.
- Understand virtual machine networking modes and safe laboratory configuration.
- Complete the assigned TryHackMe rooms and OverTheWire Bandit levels.
- Document practical work using screenshots, writeups, and a Linux command cheat sheet.

3. Cybersecurity Theory

3.1 Introduction to Cybersecurity

Cybersecurity is the practice of protecting computers, networks, applications, and data from unauthorised access, attacks, damage, and theft.

It involves preventive controls, monitoring, detection, investigation, and recovery.

3.2 CIA Triad

The CIA triad is a fundamental information security model.

Principle	Meaning	Example
Confidentiality	Information is accessible only to authorised people.	Restricting access to confidential files
Integrity	Information remains accurate and is not changed without authorisation.	Using hashes to detect file changes
Availability	Systems and information remain accessible when needed.	Maintaining backups and recovery systems

3.3 Cybersecurity Roles

Role	Main responsibility
SOC Analyst	Monitors alerts and investigates suspicious activity.
VAPT / Penetration Tester	Tests systems for vulnerabilities with authorisation.
Incident Responder	Investigates incidents, contains threats, and supports recovery.

3.4 Networking Fundamentals

I studied the purpose of IP and MAC addresses, DNS, DHCP, ports, routing, and network protocols.

I also reviewed the OSI model and TCP/IP model to understand how data travels between applications and network devices.

3.5 Linux Fundamentals

I studied the Linux file system hierarchy, command-line operations, permissions, ownership, users, groups, and process management.

3.6 Threats and Ethical Security

I learned the differences between threats, vulnerabilities, risks, and exploits. I also reviewed the importance of authorisation, testing scope, and responsible security practices.

4. Virtual Machine Lab Setup

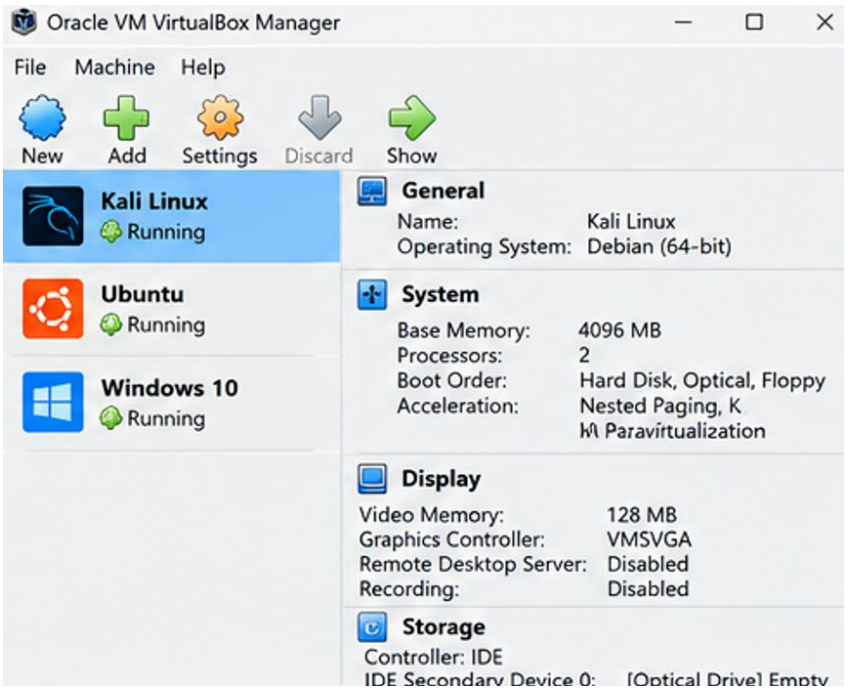
4.1 Objective

The objective was to create a controlled laboratory environment for practising operating system administration and basic networking.

4.2 Lab Components

The planned lab environment includes:

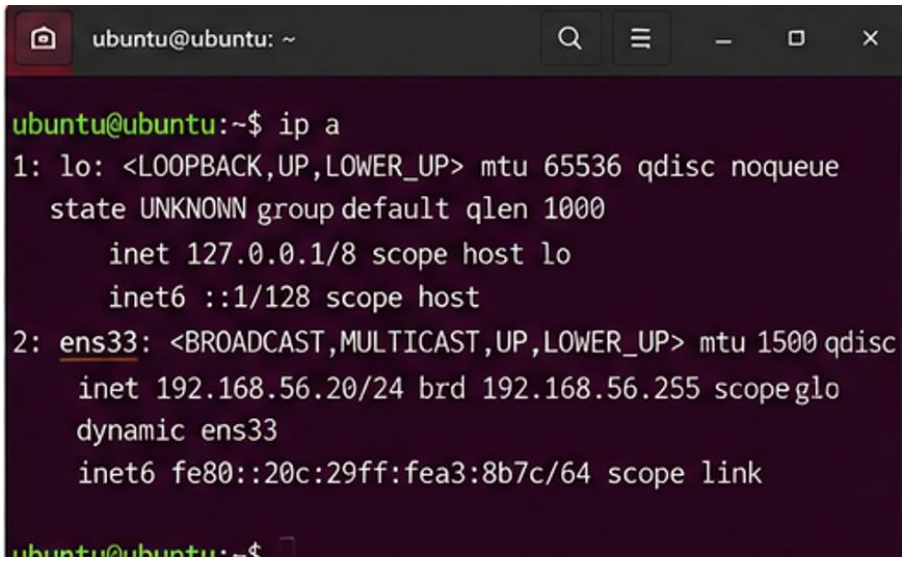
- Kali Linux — security learning and testing tools.
- Ubuntu Linux — Linux administration and command-line practice.
- Windows 10/11 — Windows operating system and connectivity practice.
- Oracle VirtualBox — virtualisation software.



4.3 Network Configuration

The lab activity involved reviewing virtual machine network settings, checking IP configurations, and testing communication between machines.

A Host-only network can provide isolated communication between the host and virtual machines. A NAT network is another option when internet access is needed for lab systems.



4.4 Connectivity Testing

The following commands were used or reviewed for network troubleshooting:

Linux:

ip a

ping -c 4 <destination-IP>

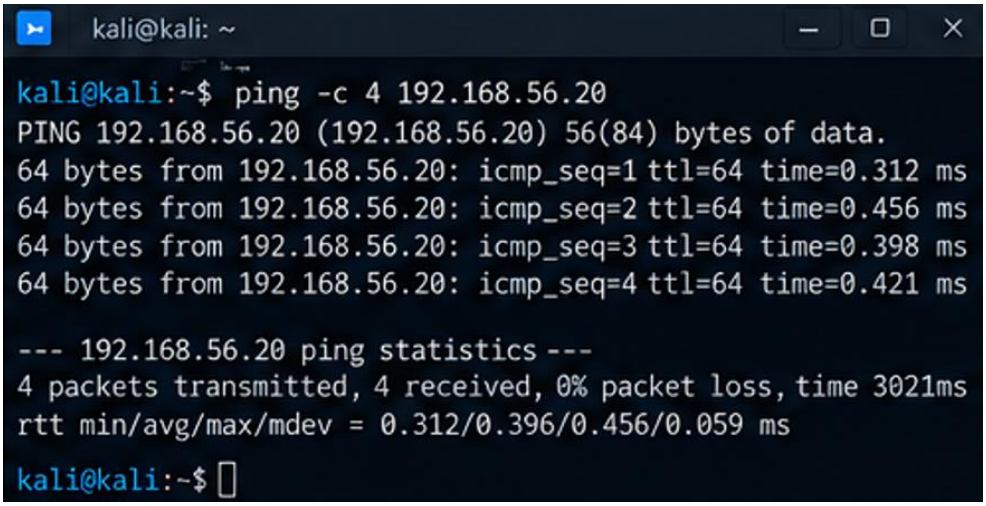
ip route

Windows:

ipconfig

ping <destination-IP>

The results should be recorded using the actual IP addresses and ping output from the lab



Evidence to insert:

- VirtualBox VM list.
- Kali Linux IP configuration.
- Ubuntu IP configuration.
- Windows IP configuration.
- Successful ping results, where verified.
- Clean snapshots of each VM, where created.
- One network diagram showing the lab architecture.

5. Linux Practical Activities

5.1 File and Directory Management

I practised navigating directories and managing files using commands such as pwd, ls, cd, mkdir, touch, cp, mv, and rm.

These commands are important for day-to-day Linux administration and laboratory work.

5.2 Searching and Text Processing

I practised using find and grep to locate files and search text. These commands are useful when analysing configuration files, logs, and unfamiliar directory structures.

5.3 Process and Network Inspection

I studied commands such as ps and ss -tulpn to inspect running processes and network sockets. Administrative privileges may be needed to display some process details.

5.4 Users, Groups, and Permissions

I practised Linux user and group administration, file ownership, and permission management.

Commands studied included:

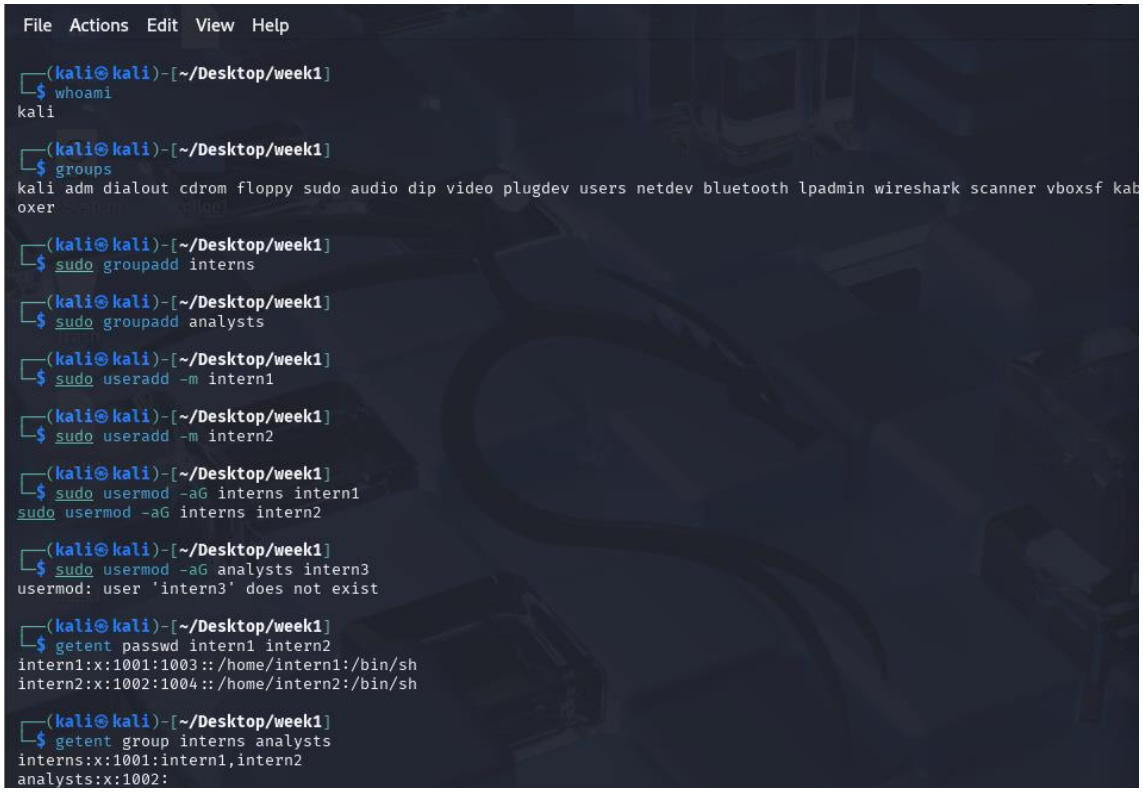
whoami

id

chmod

chown

I also reviewed how shared-directory permissions control access to files.



5.5 Cron Scheduling

I studied cron jobs, which allow commands or scripts to run automatically at scheduled times. I reviewed the cron configuration and execution evidence from my practice.

Evidence to insert:

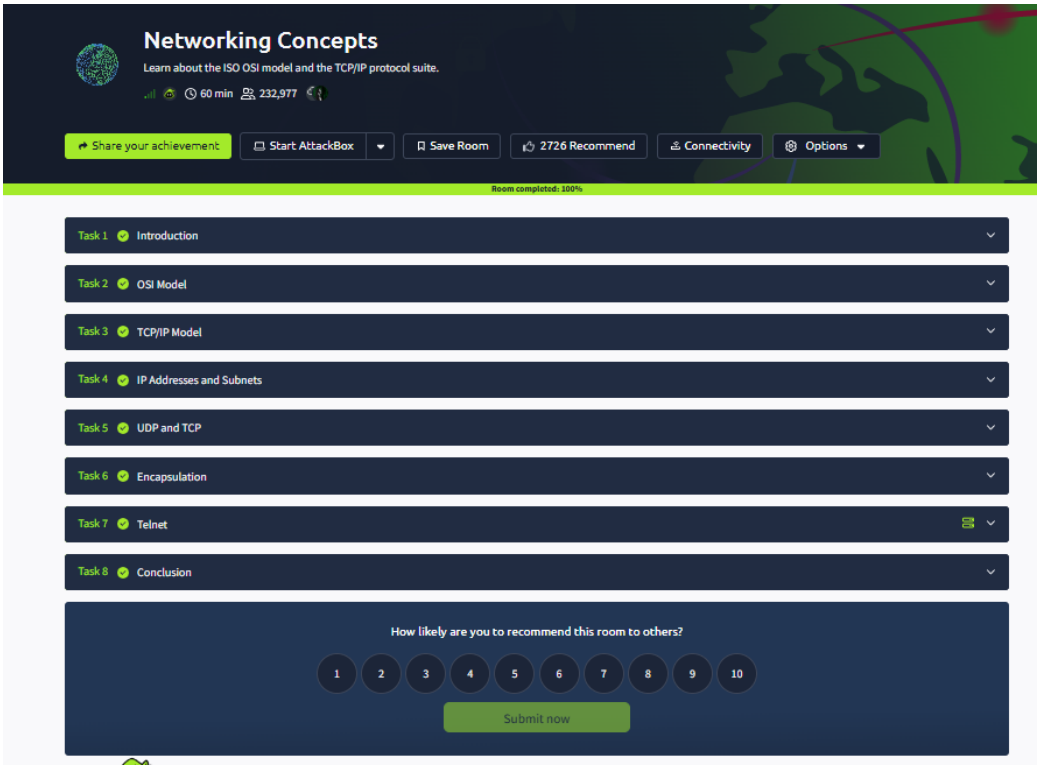
- Linux command outputs.
- User and group information.
- Shared-folder permissions.
- Process and socket inspection.
- Cron configuration and execution evidence.

6. TryHackMe Learning Activities

I worked through the assigned beginner rooms to strengthen my cybersecurity and networking fundamentals.

The rooms covered:

1. Search Skills
2. Learning Cyber Security
3. Linux Fundamentals Part 1
4. What Is Networking
5. Introductory Networking
6. Networking Concepts
7. Pyramid of Pain
8. Intro to LAN





Key Learning Outcomes

- Improved technical information-searching skills.
- Developed an understanding of common cyber threats and security roles.
- Practised basic Linux terminal concepts.
- Learned networking terminology and protocols.
- Reviewed IP addressing, ports, and LAN concepts.
- Understood how threat intelligence indicators can support security monitoring.

Evidence to insert: Actual room-completion screenshots. Refer to the separate writeup for each room for detailed notes.

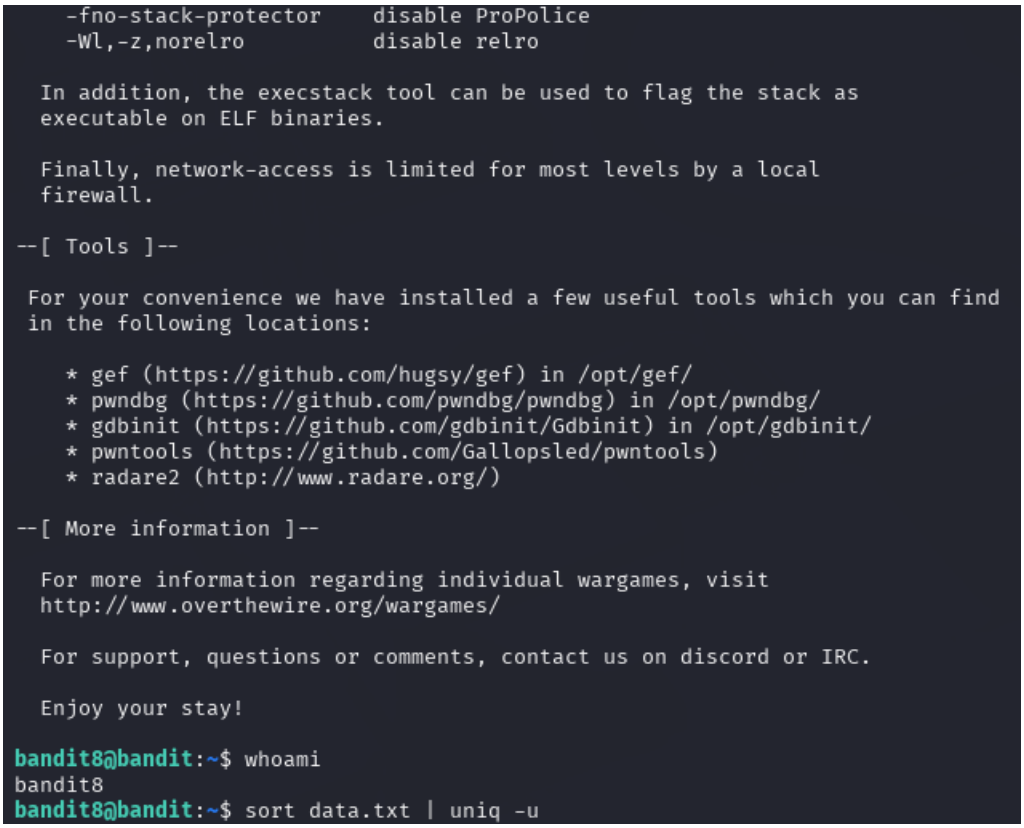
7. OverTheWire Bandit Activities

I practised beginner Linux challenges through OverTheWire Bandit, covering Levels 0–10.

The exercises introduced command-line problem-solving and file inspection.

Concepts Practised

- Connecting to a remote Linux server using SSH.
- Reading files with unusual names.
- Handling filenames containing spaces.
- Identifying hidden files.
- Inspecting file types.
- Searching by file properties.
- Using ownership and group criteria in searches.
- Searching text with grep.
- Identifying unique lines with sort and uniq.
- Extracting readable strings from data.
- Decoding Base64-encoded text.



Learning Outcomes

The exercises helped me improve my understanding of Linux commands, file handling, text processing, and basic data inspection.

Security note: Bandit passwords are excluded from the writeups and this report.

Evidence to insert: Actual level-completion screenshots and a reference to the separate Bandit writeups.

8. Tools and Technologies

Tool / Technology	Purpose
Kali Linux	Cybersecurity learning environment
Ubuntu Linux	Linux administration practice
Windows 10/11	Windows system and connectivity practice
Oracle VirtualBox	Running virtual machines
SSH	Secure remote login
Linux terminal	Command-line administration
TryHackMe	Guided cybersecurity learning
OverTheWire Bandit	Linux command-line challenges
draw.io	Network diagram creation, if used
Google Docs / Drive	Documentation and evidence storage

9. Documentation and Evidence Management

The Week 1 work is organised into separate folders for notes, writeups, screenshots, and reports.

The documentation includes theory notes, TryHackMe writeups, Bandit writeups, Linux command references, lab evidence, and the network diagram.

The purpose of this structure is to make the work easy to review and to keep practical evidence separate from explanatory documentation.

10. Challenges and Troubleshooting

During practical learning, network configuration, VM connectivity, command syntax, and file permissions required attention.

I used command-line inspection and basic connectivity tests to understand system behaviour and identify possible configuration issues.

Any unresolved issues should be recorded accurately rather than reported as successfully fixed.

11. Skills Developed

The Week 1 activities helped develop foundational knowledge in:

- Cybersecurity fundamentals and the CIA triad.
- Linux command-line operations.
- File and directory management.
- Linux permissions, ownership, and process inspection.
- Network addressing and protocol fundamentals.
- Virtual machine networking.
- Introductory security labs and technical research.
- Writing technical notes and organising evidence.

12. Conclusion

Week 1 provided a foundation in cybersecurity, Linux, and networking. The theory topics, Linux exercises, virtual machine lab activities, TryHackMe rooms, and Bandit challenges supported my understanding of basic system administration and network communication.

The practical exercises also highlighted the importance of documenting results, using secure laboratory configurations, and performing security testing only with appropriate authorisation.

These foundational skills will support my continued learning in security monitoring, vulnerability assessment, incident response, and digital forensics.

13. Supporting Documents

The following supporting materials are maintained separately:

- Eight cybersecurity theory notes.
- Linux command cheat sheet containing at least 30 entries.
- TryHackMe room writeups.
- OverTheWire Bandit writeups for Levels 0–10.
- VM lab screenshots and network diagram.
- Linux practice screenshots and command outputs.
- Weekly presentation and README file.

Report status: Review all evidence and confirm that each stated activity is supported by actual results before submitting.